

Graphical Models, the **do**-Operator, and Reinforcement Learning

Abstract

We reflect on the use of graphical models and Pearl’s **do**-operator in reinforcement learning (RL). The MDP is a causal graphical model, and the Bellman equations are **do**-calculus identities. While the standard RL literature uses conditional expectations, the **do**-operator makes the causal structure explicit and generalizes naturally to off-policy and counterfactual settings. We argue that the approach is correct, illuminating, and forward-looking.

Contents

1 Introduction

In this note we discuss the relationship between graphical models, Pearl’s **do**-operator, and reinforcement learning. The starting observation is that the MDP is a *causal graphical model*, and the Bellman equations can be derived as **do**-calculus identities. The standard RL literature uses conditional expectations instead, but the two are compatible, and the **do**-operator adds clarity in several places.

2 The MDP as a causal graphical model

The MDP trajectory

$$S_t, A_t, R_{t+1}, S_{t+1}, A_{t+1}, R_{t+2}, \dots$$

has a directed acyclic structure when unrolled in time, with edges

- $S_t \rightarrow A_t$ (policy),
- $(S_t, A_t) \rightarrow R_{t+1}$ (reward),
- $(S_t, A_t) \rightarrow S_{t+1}$ (transition),

and so on. The joint factorizes as

$$p(s_0) \prod_{k \geq 0} \pi(a_k \mid s_k) p(s_{k+1} \mid s_k, a_k) p(r_{k+1} \mid s_k, a_k, s_{k+1}).$$

This is a *causal Bayesian network* in Pearl’s sense. The Bellman equations, the value functions, and the optimality equations are all marginalization statements about this network.

3 The do-operator in RL

In Pearl's notation, $P(Y \mid \text{do}(X = x))$ is the distribution of Y when we *intervene* on X , setting it to x and cutting all incoming edges to X . This is exactly what RL needs:

- **Value of a policy** $v_\pi(s)$: intervene on $S_t = s$ and follow π thereafter:

$$v_\pi(s) = \mathbb{E}[G_t \mid \text{do}(S_t = s), A_{t+k} \sim \pi].$$

- **Action-value** $q_\pi(s, a)$: intervene on both $S_t = s$ and $A_t = a$:

$$q_\pi(s, a) = \mathbb{E}[G_t \mid \text{do}(S_t = s), \text{do}(A_t = a), A_{t+k} \sim \pi \text{ for } k \geq 1].$$

- **Optimal value** $v_*(s)$:

$$v_*(s) = \mathbb{E}[G_t \mid \text{do}(S_t = s), A_{t+k} \sim \pi_* \text{ for all } k].$$

- **Optimal action-value** $q_*(s, a)$:

$$q_*(s, a) = \mathbb{E}[G_t \mid \text{do}(S_t = s), \text{do}(A_t = a), A_{t+k} \sim \pi_* \text{ for } k \geq 1].$$

Under this reading, the Bellman equations are **do**-calculus identities:

$$v_\pi(s) = \mathbb{E}[R_{t+1} + \gamma v_\pi(S_{t+1}) \mid \text{do}(S_t = s)],$$

$$q_\pi(s, a) = \mathbb{E}[R_{t+1} + \gamma v_\pi(S_{t+1}) \mid \text{do}(S_t = s), \text{do}(A_t = a)],$$

and the optimality equations replace the policy with a max over the intervened action:

$$v_*(s) = \max_a \mathbb{E}[R_{t+1} + \gamma v_*(S_{t+1}) \mid \text{do}(S_t = s), \text{do}(A_t = a)].$$

This is exactly the structure derived by marginalization. The **do**-operator makes the *causal content* explicit: we are not conditioning on $A_t = a$ (which would require A_t to be a random variable with a distribution given S_t), we are *setting* $A_t = a$ by intervention.

4 Where the do-operator matters

4.1 Policy evaluation vs. policy improvement

- **Policy evaluation** v_π : intervene on $S_t = s$, then let the policy act. No intervention on actions. This is a conditional expectation in the graph where the policy edges are present.
- **Policy improvement**: compute $q_\pi(s, a)$ by intervening on $A_t = a$, then take the max over a . The intervention cuts the policy edge $S_t \rightarrow A_t$, so A_t is no longer distributed according to π . This is why the greedy policy is $\arg \max_a q_\pi(s, a)$: it is the best *intervention* at time t .

Without the **do**-operator, one might be tempted to write $\mathbb{E}[G_t \mid S_t = s, A_t = a]$ as a conditional expectation. That is fine when the policy is fixed and $A_t \sim \pi(\cdot \mid S_t)$, but it obscures the fact that in policy improvement we are *intervening*, not conditioning. The distinction matters conceptually, even if the numerical answer happens to coincide when $\pi(a \mid s) > 0$.

4.2 Off-policy evaluation

Off-policy evaluation is the clearest case. We have data collected under a *behavior policy* b , and we want to estimate v_π for a *target policy* π . The **do**-operator makes the problem precise:

- Data: samples from $p(S_t, A_t, R_{t+1}, S_{t+1})$ under b .
- Target: $v_\pi(s) = \mathbb{E}[G_t \mid \text{do}(S_t = s), A_{t+k} \sim \pi]$.

The gap between the two is a *causal* gap: the data has $A_t \sim b(\cdot \mid S_t)$, but we want to intervene with $A_t \sim \pi(\cdot \mid S_t)$. Importance sampling, doubly robust estimators, and the whole off-policy literature are attempts to bridge this gap. The **do**-operator gives the right language:

$$v_\pi(s) = \mathbb{E}_b \left[\frac{\pi(A_t \mid S_t)}{b(A_t \mid S_t)} G_t \mid \text{do}(S_t = s) \right],$$

where the ratio is the Radon–Nikodym derivative between the intervened and observed distributions.

4.3 Where it does not change the math

For *on-policy* evaluation, the **do**-operator and conditional expectation give the same numbers, because:

- $A_t \sim \pi(\cdot \mid S_t)$ in both cases.
- The intervention on $S_t = s$ is the same as conditioning on $S_t = s$ (since S_t has no parents in the graph at time t).

So for on-policy Bellman equations, the **do**-operator is a *conceptual clarification*, not a computational change.

5 Why RL was developed without the do-operator

Historically:

- **Bellman (1957)** formulated dynamic programming in terms of conditional expectations and policy operators. The MDP was a stochastic process, not a causal model.
- **Pearl (1980s–1990s)** developed graphical models and the **do**-calculus, but the focus was on causal inference from observational data, not on sequential decision-making.
- **Sutton & Barto (1998, 2018)** wrote the standard RL textbook in the Bellman tradition, using $p(s', r \mid s, a)$ and conditional expectations. The **do**-operator is not mentioned.
- **Modern causal RL** (Bareinboim, Zhang, and others, 2020s) has begun to connect the two, especially for off-policy evaluation and counterfactual reasoning.

So RL was formulated in the conditional-expectation language, and the **do**-operator was not part of the vocabulary. But the two are compatible, and the **do**-operator adds clarity in several places.

6 What the do-operator adds to RL

Aspect	Without do-operator	With do-operator
Policy evaluation	$\mathbb{E}[G_t \mid S_t = s]$	$\mathbb{E}[G_t \mid \text{do}(S_t = s), A \sim \pi]$
Action-value	$\mathbb{E}[G_t \mid S_t = s, A_t = a]$	$\mathbb{E}[G_t \mid \text{do}(S_t = s), \text{do}(A_t = a), A_{t+1} \sim \pi]$
Greedy policy	$\arg \max_a q_\pi(s, a)$	$\arg \max_a \mathbb{E}[G_t \mid \text{do}(S_t = s), \text{do}(A_t = a)]$
Off-policy	Importance sampling (ad hoc)	Radon–Nikodym derivative
Counterfactuals	Not expressible	$\mathbb{E}[G_t \mid \text{do}(A_t = a'), A_t^{\text{obs}} = a]$
Causal confusion	Possible	Explicit

The *counterfactual* case is where the **do**-operator is genuinely indispensable. In RL, we often want to ask: “What would have happened if the agent had taken action a' instead of the observed action a ?” This is a counterfactual query, and it cannot be expressed with conditional expectations alone. The **do**-operator handles it directly:

$$\mathbb{E}[G_t \mid \text{do}(A_t = a'), A_t^{\text{obs}} = a, S_t = s].$$

This is the basis of counterfactual reasoning in RL, with applications in:

- credit assignment (which action caused the reward?),
- counterfactual regret minimization (used in poker and other games),
- causal imitation learning (learning from demonstrations with hidden confounders),
- robust RL (what if the environment changes?).

7 Is the approach non-standard?

Yes and no.

- **Non-standard in presentation:** Most RL textbooks and courses derive Bellman equations with conditional expectations, not **do**-calculus. The approach is unusual in that sense.
- **Standard in content:** The equations derived are the same as the standard ones. The **do**-operator does not change the numerical answers for on-policy evaluation; it only makes the causal structure explicit.
- **Forward-looking:** As RL increasingly intersects with causal inference (off-policy, counterfactuals, robustness, imitation), the **do**-operator will become more standard.

8 Assessment

The approach is correct, illuminating, and pedagogically valuable. Here is why:

1. It makes the causal assumptions explicit. The MDP assumes that the state is a sufficient statistic for the future (Markov property) and that actions causally affect rewards and next states. The **do**-operator makes these assumptions precise.

2. It clarifies the difference between conditioning and intervening. In policy improvement, we intervene on the action. In policy evaluation, we do not.
3. It generalizes to off-policy and counterfactual settings. The standard conditional-expectation language is awkward for these; the **do**-operator is natural.
4. It connects RL to the broader causal inference literature. This is increasingly important as the fields converge.

The main caveat is that the **do**-operator is *not necessary* for the standard on-policy Bellman equations. One can derive them with conditional expectations and get the same results. The **do**-operator adds conceptual clarity, not computational power, for the basic case. But for off-policy, counterfactual, and causal RL, the **do**-operator is the right tool, and the field is moving in that direction.

9 A suggestion for a rigorous derivation

To make the approach rigorous and publishable:

1. **State the causal graph explicitly** (the unrolled MDP DAG with policy, transition, and reward edges).
2. **Define the do-operator** in this graph, distinguishing interventions on S_t (fixing the state) from interventions on A_t (fixing the action).
3. **Derive the Bellman equations** as **do**-calculus identities, using the truncated factorization formula (Pearl’s Theorem 3.2.2 or the g-formula).
4. **Show the equivalence** with the standard conditional-expectation derivation for on-policy evaluation.
5. **Extend to off-policy** by introducing the Radon–Nikodym derivative.
6. **Extend to counterfactuals** by introducing the twin-network representation (Pearl’s counterfactual graph).

This would be a contribution: a causal graphical model treatment of RL, bridging the Bellman tradition and the Pearl tradition. There are papers on causal RL (Bareinboim, Zhang, and others) that go in this direction, but a definitive reference that does exactly this is not known to the author.

10 Summary

Question	Answer
Is the approach legitimate?	Yes. The MDP is a causal graphical model; Bellman equations are do -calculus identities.
Is it standard?	Not in presentation, but the content is equivalent for on-policy evaluation.
Does the do -operator change the math?	No for on-policy; yes conceptually for policy improvement, off-policy, and counterfactuals.
Why was it not used historically?	RL was developed in the Bellman tradition (conditional expectations); the do -operator came from a different community.
Is it worth pursuing?	Yes. It clarifies causal assumptions and generalizes to off-policy and counterfactual settings.

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